

ARPad: Compound Spatial-Semantic Gesture Interaction in Augmented Reality

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ABSTRACT

In this paper, we propose the concept of compound spatial-semantic gesture interaction, in which spatial semantics is defined as the combination of world-level spatial semantics and object-level spatial semantics. The semantics of compound spatial-semantic gesture is determined by both the gesture and the spatial location of the gesture, which can increase the interaction bandwidth of gestures and improve the user experience. We present three typical compound spatial-semantic gesture application scenarios. Finally, compound spatial-semantic concepts can be applied to other interaction channels, which provides a feasible and general implementation method for expanding the interaction in Augmented Reality (AR) scenarios.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—Interaction techniques—Gestural input

1 INTRODUCTION

Real-time interaction with virtual content is an important research issue in most AR experiences. Gesture interaction with simple and intuitive characteristics has become an extensively used interaction method in AR Scenes [4, 6]. The interaction space in AR scenes extends to the entire physical space. However, existing interactions [2, 3] often lack effective utilization of the contextual space for interaction actions, increasing the difficulty and complexity of interaction interface design and increasing the learning and usage costs of users, thus affecting AR interaction efficiency and user experience.

Based on the wide physical space activity range and various control objects, the semantics of interaction actions in AR applications often need to be associated with specific physical space locations and physical control objects. In this paper, we propose compound spatial semantics by combining specific interaction actions and spatial location information, giving different semantic information to the same interaction action so as to enhance the interaction space and reduce memory and usage cost. Moreover, we present the compound spatial-semantic gesture interaction, which applies compound spatial semantics to the traditional gesture interaction in AR. Finally, we discuss the application scenarios of compound spatial-semantic gestures, which shows that compound spatial-semantic gestures have certain advantages.

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2 COMPOUND SPATIAL SEMANTICS

In terms of ordinary interactions, for an interaction command cmd , the interaction semantics s is the same at any spatial location sp based on the general rule function f_1 .

$$s = f_1(cmd) \quad (1)$$

Since some prior research [1, 5] on proxemic interaction has existed, they do not effectively use the spatial location context. We add the new interaction channel by sufficiently utilizing the location information in physical space so that the interaction command contains compound spatial semantics. Unlike ordinary interactions, the same interaction command cmd will generate different semantics s in different spatial locations sp based on the compound spatial-semantic rule function f_2 .

$$s = f_2(cmd, sp) \quad (2)$$

In compound spatial semantics, spatial location information is divided into world spatial information and object spatial information based on different coordinate systems, both of which contain specific semantics. The world-level spatial semantics is mainly based on the absolute spatial location information in the World Coordinate System, a reference system with the invariant origin and axis direction. Object-level spatial semantics is mainly based on the relative spatial location information in the Object Coordinate System, a reference system established by the location and pose of the physical object in reality.

Based on the morphological characteristics of interaction objects, the object space can be described and divided by the Cartesian coordinate system, spherical coordinate system, and cylindrical coordinate system. The application scenarios in this paper mainly use the horizontal plane, so we use a cubic object space based on the Cartesian coordinate system.

In the unified coordinate space, world space and object space are nested. We take world space as a container in which multiple object spaces are nested, and even different object spaces can be nested within each other. Therefore, the same interaction command may be in multiple object spaces simultaneously and have different functional semantics in different spaces. We set the priority for each object space and select the object-level spatial semantics of the interaction command through the priority of the object space. According to the different object spaces where the interaction command is located, there are three cases: in single object space (Fig. 1a), in multiple object spaces (Fig. 1b), and not in the object space (Fig. 1c).

3 COMPOUND SPATIAL-SEMANTIC GESTURE INTERACTION

We apply compound spatial semantics to gesture interaction in AR scenes and design the related interaction as well as processing procedures. The compound spatial-semantic gesture divides object space into surface space and near-surface space. Touch gestures that use surface space information require touching a physical object; in-air

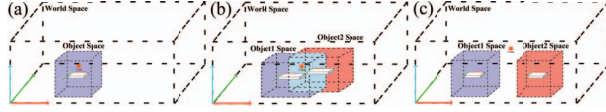


Figure 1: The different object spaces where the interaction command is located. (a) The interaction instruction in a single object space; (b) the interaction instruction in multiple object Spaces; (c) the interaction instruction not in any object space.

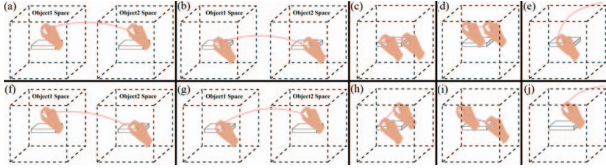


Figure 2: The types of compound spatial-semantic gestures.

gestures that use near-surface space information do not require touching a physical object but utilize the contextual space information of the physical object.

The semantics of compound spatial-semantic gesture is determined by both the gesture and the spatial location of the gesture, leading to the difference in the process of ordinary gestures. Fig. 2 shows the types of compound spatial-semantic gestures. Compound spatial-semantic gestures can be roughly divided into three cases: from object1 space to object2 space (Fig. 1a, Fig. 1b, Fig. 1f, Fig. 1g), single object space (Fig. 1c, Fig. 1d, Fig. 1h, Fig. 1i), and from world space to object space (Fig. 1e, Fig. 1j). We increase the interaction input channel by giving the gesture spatial semantic information, which not only makes use of the characteristics of the gesture itself but also enhances the use of gesture space context. Since gesture information and spatial location information are used to encode and map interaction semantics simultaneously, the compound spatial-semantic gesture can increase the interaction bandwidth of gestures and improve the efficiency of gesture interaction and user experience.

4 APPLICATION SCENARIOS

Target selection. In AR office scenarios, the selection of multiple virtual screens is a common problem. In the object space of each virtual screen, we set the gesture in the surface space of the virtual screen to be multi-selected, the gesture in the near-surface space to be single-selected, and the gesture outside the object space of all virtual screens to be fully selected.

Physical object manipulation. In daily life, it is usually necessary to manipulate physical objects quickly. We control the properties of notebook computers and desk lamps based on the moving direction and distance of gestures in object space. The horizontal movement of the gesture in the surface space indicates the brightness adjustment of the screen, the vertical movement of the gesture in the surface space indicates the volume adjustment, and the up and down movement of the gesture in the near-surface space indicates the brightness control of the lamp.

File operation. When working on a computer, manipulation of documents is usually required. We take object1 space as the starting point and object2 space as the endpoint. Among the object1 space, the gesture in the surface space represents single-selection, and the gesture in the near-surface space represents multi-selection. Among the object2 space, the gesture in the surface space represents copying and pasting, and the gesture in the near-surface space represents movement and pasting. Therefore, four specific interaction semantics are designed based on the positional relationship between

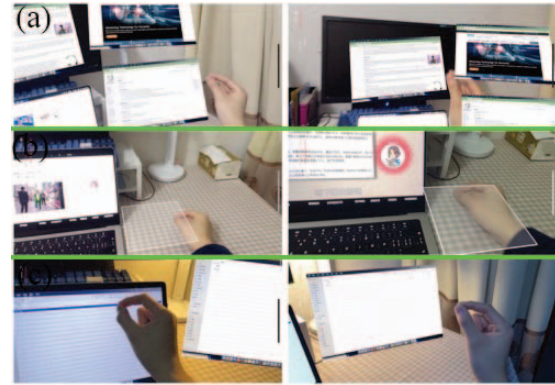


Figure 3: Three compound spatial-semantic gesture application scenarios. (a) target selection; (b) physical object manipulation; (c) document operation.

surface space and near-surface space.

5 CONCLUSIONS AND FUTURE WORK

This paper proposes the concept of compound spatial semantics, which combines the specific interaction action with the spatial location information to give the interaction action spatial semantic information. Moreover, applying compound spatial semantics to gesture interaction in AR, we propose the compound spatial-semantic gesture. Finally, to explore the capabilities of the compound spatial-semantic gestures, we show three application scenarios: target selection, physical object manipulation, and file operation. In the future, we will supplement relevant user studies to verify the execution efficiency and usability of compound spatial semantic gestures. Meanwhile, we will study different object space morphologies.

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